BRACT's

Vishwakarma Institute of Technology, Pune

Practical Implementation Sheet

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Division: C Batch: 2

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Subject Name: Deep Learning

Problem Statement:

Implement transfer learning using pre-trained AlexNet, VGG16, ResNet50, and EfficientNetB0 models for image classification, and compare their performance.

Algorithm :

Introduction

Transfer learning is a deep learning technique in which a model that has already been trained on a large dataset is reused for a new but related task.
Instead of training a deep neural network from the beginning, a pretrained model can be used as a feature extractor. The pretrained layers have already learned useful features such as:

- Edges
- Corners
- Textures
- Shapes
- Patterns
- Object structures

These learned features can then be used for classification on a new dataset.
In this experiment, the CIFAR-10 dataset is used for image classification. Four popular CNN architectures are compared:

1. AlexNet
2. VGG16
3. ResNet50

4. EfficientNetB0

CIFAR-10 Dataset

CIFAR-10 is a commonly used image classification dataset containing 60,000 color images belonging to 10 classes.

Each original image has a resolution of:

$32 \times 32 \times 3$

The ten classes are:

Class	Description
0	Airplane
1	Automobile
2	Bird
3	Cat
4	Deer
5	Dog
6	Frog
7	Horse
8	Ship
9	Truck

For this experiment, a smaller subset is used to reduce computational and memory requirements:

- Training samples: 5000
- Validation samples: 1000
- Test samples: 1000

The CIFAR-10 images are resized from 32×32 to 224×224, because VGG16, ResNet50 and the chosen AlexNet architecture operate with larger image inputs.

Models Used

AlexNet

AlexNet is a convolutional neural network introduced by Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton.

Its architecture contains:

- Convolutional layers
- ReLU activation
- Max-pooling layers
- Fully connected layers
- Dropout

AlexNet played an important role in demonstrating the effectiveness of deep CNNs for image classification.

VGG16

VGG16 is a deep convolutional neural network developed by the Visual Geometry Group at the University of Oxford.

It contains:

- 13 convolutional layers

- 3 fully connected layers
- 16 weight layers in total

VGG16 primarily uses small:
3 × 3 convolution filters
The pretrained VGG16 model used in this experiment has been trained on ImageNet.
The convolutional base is frozen, meaning its pretrained weights are not changed during initial training.

6. ResNet50
ResNet50 is a deep CNN architecture based on residual learning.
It contains approximately 50 layers.
The major innovation of ResNet is the residual/skip connection.
Instead of directly learning:
a residual block learns:
and then produces:
Skip connections help reduce the vanishing-gradient problem and allow very deep neural networks to be trained effectively.
In this experiment, ResNet50 uses ImageNet pretrained weights.

7. EfficientNetB0
EfficientNet is a family of CNN models designed to provide a good balance between:

- Accuracy
- Number of parameters
- Computational cost

EfficientNet uses a technique called compound scaling, which scales:

- Network depth
- Network width
- Image resolution

in a coordinated way.
EfficientNetB0 is the baseline model of the EfficientNet family.
It is generally much more parameter-efficient than older architectures while providing strong classification performance.

Evaluation Metrics
The models are compared using several metrics.
Accuracy
Accuracy represents the percentage of correctly classified images.

$$Accuracy = \frac{Correct\ Predictions}{Total\ Predictions}$$

Precision
Precision measures how many predicted positive samples are actually correct.

$$Precision = \frac{TP}{TP + FP}$$

Recall
Recall measures how many actual positive samples are correctly identified.

$$Recall = \frac{TP}{TP + FN}$$

F1-score
F1-score combines precision and recall.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Code :

```
# TRANSFER LEARNING USING PRETRAINED:
# AlexNet, VGG16, ResNet50, EfficientNetB0
# Dataset: CIFAR-10

import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import DataLoader, Subset
import torchvision
import torchvision.transforms as transforms
from torchvision import models
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import time
import copy
import gc
from sklearn.model_selection import train_test_split
from sklearn.metrics import confusion_matrix, classification_report


device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print("Using device:", device)

IMG_SIZE = 224
BATCH_SIZE = 16
EPOCHS = 3
NUM_CLASSES = 10
TRAIN_SAMPLES = 5000
TEST_SAMPLES = 1000
RANDOM_STATE = 42
LEARNING_RATE = 0.001

class_names = [
```

```
    "airplane", "automobile", "bird", "cat", "deer",
    "dog", "frog", "horse", "ship", "truck"
]


# ImageNet pretrained models expect normalized RGB images.
train_transform = transforms.Compose([
    transforms.Resize((IMG_SIZE, IMG_SIZE)),
    transforms.RandomHorizontalFlip(),
    transforms.ToTensor(),
    transforms.Normalize(
        mean=[0.485, 0.456, 0.406],
        std=[0.229, 0.224, 0.225]
    )
])

test_transform = transforms.Compose([
    transforms.Resize((IMG_SIZE, IMG_SIZE)),
    transforms.ToTensor(),
    transforms.Normalize(
        mean=[0.485, 0.456, 0.406],
        std=[0.229, 0.224, 0.225]
    )
])

# -------------------- LOAD CIFAR-10 --------------------
full_train_dataset = torchvision.datasets.CIFAR10(
    root="./data",
    train=True,
    download=True,
    transform=train_transform
)

# Separate copy without augmentation for validation
full_train_dataset_no_aug = torchvision.datasets.CIFAR10(
    root="./data",
    train=True,
    download=True,
    transform=test_transform
)

full_test_dataset = torchvision.datasets.CIFAR10(
    root="./data",
    train=False,
    download=True,
```

```
    transform=test_transform
)


# -------------------- TRAIN / VALIDATION SPLIT --------------------
train_indices = np.arange(TRAIN_SAMPLES)
test_indices = np.arange(TEST_SAMPLES)

all_labels = np.array(full_train_dataset.targets)[train_indices]

train_idx, val_idx = train_test_split(
    train_indices,
    test_size=0.20,
    random_state=RANDOM_STATE,
    stratify=all_labels
)

train_dataset = Subset(full_train_dataset, train_idx)
val_dataset = Subset(full_train_dataset_no_aug, val_idx)
test_dataset = Subset(full_test_dataset, test_indices)

train_loader = DataLoader(
    train_dataset,
    batch_size=BATCH_SIZE,
    shuffle=True,
    num_workers=0
)

val_loader = DataLoader(
    val_dataset,
    batch_size=BATCH_SIZE,
    shuffle=False,
    num_workers=0
)

test_loader = DataLoader(
    test_dataset,
    batch_size=BATCH_SIZE,
    shuffle=False,
    num_workers=0
)

print("Training images:", len(train_dataset))
print("Validation images:", len(val_dataset))
print("Testing images:", len(test_dataset))
```

```python
# -------------------- LOAD PRETRAINED MODELS --------------------
def get_pretrained_model(model_name):

    if model_name == "AlexNet":
        print("\nLoading pretrained AlexNet...")
        model = models.alexnet(
            weights=models.AlexNet_Weights.DEFAULT
        )

        # Freeze all pretrained parameters
        for param in model.parameters():
            param.requires_grad = False

        # Replace ImageNet 1000-class output with CIFAR-10 10-class output
        input_features = model.classifier[6].in_features
        model.classifier[6] = nn.Linear(
            input_features, NUM_CLASSES
        )

    elif model_name == "VGG16":
        print("\nLoading pretrained VGG16...")
        model = models.vgg16(
            weights=models.VGG16_Weights.DEFAULT
        )

        # Freeze all pretrained parameters
        for param in model.parameters():
            param.requires_grad = False

        input_features = model.classifier[6].in_features
        model.classifier[6] = nn.Linear(
            input_features, NUM_CLASSES
        )

    elif model_name == "ResNet50":
        print("\nLoading pretrained ResNet50...")
        model = models.resnet50(
            weights=models.ResNet50_Weights.DEFAULT
        )

        # Freeze all pretrained parameters
        for param in model.parameters():
            param.requires_grad = False
```

```
        input_features = model.fc.in_features
        model.fc = nn.Linear(
            input_features, NUM_CLASSES
        )

    elif model_name == "EfficientNetB0":
        print("\nLoading pretrained EfficientNetB0...")
        model = models.efficientnet_b0(
            weights=models.EfficientNet_B0_Weights.DEFAULT
        )

        # Freeze all pretrained parameters
        for param in model.parameters():
            param.requires_grad = False

        input_features = model.classifier[1].in_features
        model.classifier[1] = nn.Linear(
            input_features, NUM_CLASSES
        )

    else:
        raise ValueError("Unknown model name")

    return model.to(device)


# -------------------- TRAINING FUNCTION --------------------
def train_model(model, train_loader, val_loader, epochs):

    criterion = nn.CrossEntropyLoss()

    # Only the newly added classifier layer is trainable
    optimizer = optim.Adam(
        filter(lambda p: p.requires_grad, model.parameters()),
        lr=LEARNING_RATE
    )

    history = {
        "train_loss": [],
        "train_accuracy": [],
        "val_loss": [],
        "val_accuracy": []
    }
```

```
    best_val_accuracy = 0.0
    best_weights = copy.deepcopy(model.state_dict())

    start_time = time.time()

    for epoch in range(epochs):

        print(f"\nEpoch {epoch + 1}/{epochs}")

        # ---------------- TRAIN ----------------
        model.train()

        running_loss = 0.0
        correct = 0
        total = 0

        for images, labels in train_loader:

            images = images.to(device)
            labels = labels.to(device)

            optimizer.zero_grad()

            outputs = model(images)
            loss = criterion(outputs, labels)

            loss.backward()
            optimizer.step()

            running_loss += loss.item() * images.size(0)

            _, predicted = torch.max(outputs, 1)

            total += labels.size(0)
            correct += (predicted == labels).sum().item()

        train_loss = running_loss / total
        train_accuracy = correct / total

        # ---------------- VALIDATION ----------------
        model.eval()

        val_loss_total = 0.0
        val_correct = 0
        val_total = 0
```

```
        with torch.no_grad():

            for images, labels in val_loader:

                images = images.to(device)
                labels = labels.to(device)

                outputs = model(images)
                loss = criterion(outputs, labels)

                val_loss_total += loss.item() * images.size(0)

                _, predicted = torch.max(outputs, 1)

                val_total += labels.size(0)
                val_correct += (predicted == labels).sum().item()

        val_loss = val_loss_total / val_total
        val_accuracy = val_correct / val_total

        history["train_loss"].append(train_loss)
        history["train_accuracy"].append(train_accuracy)
        history["val_loss"].append(val_loss)
        history["val_accuracy"].append(val_accuracy)

        if val_accuracy > best_val_accuracy:
            best_val_accuracy = val_accuracy
            best_weights = copy.deepcopy(model.state_dict())

        print(
            f"Train Loss: {train_loss:.4f} | "
            f"Train Accuracy: {train_accuracy * 100:.2f}%"
        )
        print(
            f"Val Loss: {val_loss:.4f} | "
            f"Val Accuracy: {val_accuracy * 100:.2f}%"
        )

    training_time = time.time() - start_time

    # Restore best validation model
    model.load_state_dict(best_weights)

    return model, history, training_time
```

```python
# -------------------- TEST FUNCTION --------------------
def evaluate_model(model, test_loader):

    model.eval()

    criterion = nn.CrossEntropyLoss()

    total_loss = 0.0
    correct = 0
    total = 0

    all_labels = []
    all_predictions = []

    with torch.no_grad():

        for images, labels in test_loader:

            images = images.to(device)
            labels = labels.to(device)

            outputs = model(images)
            loss = criterion(outputs, labels)

            total_loss += loss.item() * images.size(0)

            _, predicted = torch.max(outputs, 1)

            total += labels.size(0)
            correct += (predicted == labels).sum().item()

            all_labels.extend(labels.cpu().numpy())
            all_predictions.extend(predicted.cpu().numpy())

    accuracy = correct / total
    test_loss = total_loss / total

    return (
        accuracy,
        test_loss,
        np.array(all_labels),
        np.array(all_predictions)
    )
```

```python
# -------------------- RUN ALL FOUR MODELS --------------------
model_names = [
    "AlexNet",
    "VGG16",
    "ResNet50",
    "EfficientNetB0"
]

results = []
histories = {}
true_labels = {}
predictions = {}

for model_name in model_names:

    print("\n" + "=" * 70)
    print("MODEL:", model_name)
    print("=" * 70)

    model = get_pretrained_model(model_name)

    model, history, training_time = train_model(
        model,
        train_loader,
        val_loader,
        EPOCHS
    )

    accuracy, test_loss, y_true, y_pred = evaluate_model(
        model,
        test_loader
    )

    histories[model_name] = history
    true_labels[model_name] = y_true
    predictions[model_name] = y_pred

    results.append({
        "Model": model_name,
        "Test Accuracy": accuracy,
        "Test Loss": test_loss,
        "Training Time (sec)": training_time
    })
```

```
    print("\nTest Accuracy:", f"{accuracy * 100:.2f}%")
    print("Test Loss:", f"{test_loss:.4f}")
    print("Training Time:", f"{training_time:.2f} seconds")

    # Free memory before loading next model
    del model
    gc.collect()

    if torch.cuda.is_available():
        torch.cuda.empty_cache()


# -------------------- RESULTS TABLE --------------------
results_df = pd.DataFrame(results)

results_df["Accuracy (%)"] = (
    results_df["Test Accuracy"] * 100
)

print("\n" + "=" * 70)
print("FINAL MODEL COMPARISON")
print("=" * 70)

print(
    results_df[
        [
            "Model",
            "Accuracy (%)",
            "Test Loss",
            "Training Time (sec)"
        ]
    ].round(4).to_string(index=False)
)


# -------------------- BEST MODEL --------------------
best_row = results_df.loc[
    results_df["Test Accuracy"].idxmax()
]

print("\nBest Model:", best_row["Model"])
print(
    "Best Accuracy:",
    f"{best_row['Accuracy (%)']:.2f}%"
```

```
)


# -------------------- ACCURACY GRAPH --------------------
plt.figure(figsize=(10, 6))

plt.bar(
    results_df["Model"],
    results_df["Accuracy (%)"]
)

plt.title("Transfer Learning Model Accuracy Comparison")
plt.xlabel("Model")
plt.ylabel("Test Accuracy (%)")
plt.ylim(0, 100)
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()


# -------------------- TRAINING TIME GRAPH --------------------
plt.figure(figsize=(10, 6))

plt.bar(
    results_df["Model"],
    results_df["Training Time (sec)"]
)

plt.title("Training Time Comparison")
plt.xlabel("Model")
plt.ylabel("Training Time (seconds)")
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()


# -------------------- ACCURACY CURVES --------------------
for model_name in model_names:

    history = histories[model_name]

    epochs_range = range(
        1,
        len(history["train_accuracy"]) + 1
    )
```

```
    plt.figure(figsize=(8, 5))

    plt.plot(
        epochs_range,
        history["train_accuracy"],
        marker="o",
        label="Training Accuracy"
    )

    plt.plot(
        epochs_range,
        history["val_accuracy"],
        marker="o",
        label="Validation Accuracy"
    )

    plt.title(
        model_name + " - Training vs Validation Accuracy"
    )

    plt.xlabel("Epoch")
    plt.ylabel("Accuracy")
    plt.legend()
    plt.grid(True)
    plt.tight_layout()
    plt.show()


# -------------------- LOSS CURVES --------------------
for model_name in model_names:

    history = histories[model_name]

    epochs_range = range(
        1,
        len(history["train_loss"]) + 1
    )

    plt.figure(figsize=(8, 5))

    plt.plot(
        epochs_range,
        history["train_loss"],
        marker="o",
```

```
        label="Training Loss"
    )

    plt.plot(
        epochs_range,
        history["val_loss"],
        marker="o",
        label="Validation Loss"
    )

    plt.title(
        model_name + " - Training vs Validation Loss"
    )

    plt.xlabel("Epoch")
    plt.ylabel("Loss")
    plt.legend()
    plt.grid(True)
    plt.tight_layout()
    plt.show()


# --------------------- CONFUSION MATRICES ---------------------
for model_name in model_names:

    y_true = true_labels[model_name]
    y_pred = predictions[model_name]

    cm = confusion_matrix(
        y_true,
        y_pred
    )

    plt.figure(figsize=(10, 8))

    sns.heatmap(
        cm,
        annot=True,
        fmt="d",
        cmap="Blues",
        xticklabels=class_names,
        yticklabels=class_names
    )

    plt.title(
```

```
        model_name + " - Confusion Matrix"
    )

    plt.xlabel("Predicted Class")
    plt.ylabel("Actual Class")
    plt.tight_layout()
    plt.show()

# -------------------- CLASSIFICATION REPORTS --------------------
for model_name in model_names:

    print("\n" + "=" * 70)
    print("CLASSIFICATION REPORT:", model_name)
    print("=" * 70)

    print(
        classification_report(
            true_labels[model_name],
            predictions[model_name],
            target_names=class_names
        )
    )


# -------------------- FINAL CONCLUSION --------------------
best_accuracy_model = results_df.loc[
    results_df["Test Accuracy"].idxmax(),
    "Model"
]

fastest_model = results_df.loc[
    results_df["Training Time (sec)"].idxmin(),
    "Model"
]

print("\n" + "=" * 80)
print("CONCLUSION")
print("=" * 80)

print(
    "Best performing model:",
    best_accuracy_model
)

print(
```

```
    "Fastest model:",
    fastest_model
)
```

Output :

Using device: cuda

100%|| 170M/170M [01:42<00:00, 1.66MB/s]

Training images: 4000

Validation images: 1000

Testing images: 1000

MODEL: AlexNet

==========

Loading pretrained AlexNet...

Downloading: "https://download.pytorch.org/models/alexnet-owt-7be5be79.pth" to /root/.cache/torch/hub/checkpoints/alexnet-owt-7be5be79.pth

100%|| 233M/233M [00:01<00:00, 175MB/s]

Epoch 1/3

Train Loss: 0.9454 | Train Accuracy: 66.77%

Val Loss: 0.7780 | Val Accuracy: 74.40%

Epoch 2/3

Train Loss: 0.6774 | Train Accuracy: 76.92%

Val Loss: 0.6886 | Val Accuracy: 76.60%

Epoch 3/3

Train Loss: 0.5571 | Train Accuracy: 80.15%

Val Loss: 0.6859 | Val Accuracy: 78.30%

Test Accuracy: 76.20%

Test Loss: 0.7298

Training Time: 29.85 seconds

MODEL: VGG16

==========

Loading pretrained VGG16...

Downloading: "https://download.pytorch.org/models/vgg16-397923af.pth" to /root/.cache/torch/hub/checkpoints/vgg16-397923af.pth

100%|██████████| 528M/528M [00:06<00:00, 83.8MB/s]

Epoch 1/3

Train Loss: 0.8553 | Train Accuracy: 70.28%

Val Loss: 0.6057 | Val Accuracy: 78.80%

Epoch 2/3

Train Loss: 0.6157 | Train Accuracy: 78.00%

Val Loss: 0.5659 | Val Accuracy: 81.80%

Epoch 3/3

Train Loss: 0.5511 | Train Accuracy: 80.47%

Val Loss: 0.5760 | Val Accuracy: 81.90%

Test Accuracy: 78.70%

Test Loss: 0.5900

Training Time: 89.80 seconds

MODEL: ResNet50

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Loading pretrained ResNet50...

Downloading: "https://download.pytorch.org/models/resnet50-11ad3fa6.pth" to /root/.cache/torch/hub/checkpoints/resnet50-11ad3fa6.pth

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Epoch 1/3

Train Loss: 1.3753 | Train Accuracy: 58.85%

Val Loss: 0.9361 | Val Accuracy: 71.40%

Epoch 2/3

Train Loss: 0.8977 | Train Accuracy: 72.05%

Val Loss: 0.7807 | Val Accuracy: 75.60%

Epoch 3/3

Train Loss: 0.7679 | Train Accuracy: 75.38%

Val Loss: 0.7107 | Val Accuracy: 77.40%

Test Accuracy: 75.60%

Test Loss: 0.7445

Training Time: 70.72 seconds

MODEL: EfficientNetB0

==========

Loading pretrained EfficientNetB0...

Downloading: "https://download.pytorch.org/models/efficientnet_b0_rwightman-7f5810bc.pth" to /root/.cache/torch/hub/checkpoints/efficientnet_b0_rwightman-7f5810bc.pth

100%|| 20.5M/20.5M [00:00<00:00, 158MB/s]

Epoch 1/3

Train Loss: 1.4657 | Train Accuracy: 56.33%

Val Loss: 0.9876 | Val Accuracy: 73.50%

Epoch 2/3

Train Loss: 0.9843 | Train Accuracy: 69.33%

Val Loss: 0.8094 | Val Accuracy: 75.30%

Epoch 3/3

Train Loss: 0.8712 | Train Accuracy: 72.55%

Val Loss: 0.7612 | Val Accuracy: 76.80%

Test Accuracy: 76.80%

Test Loss: 0.7526

Training Time: 45.11 seconds

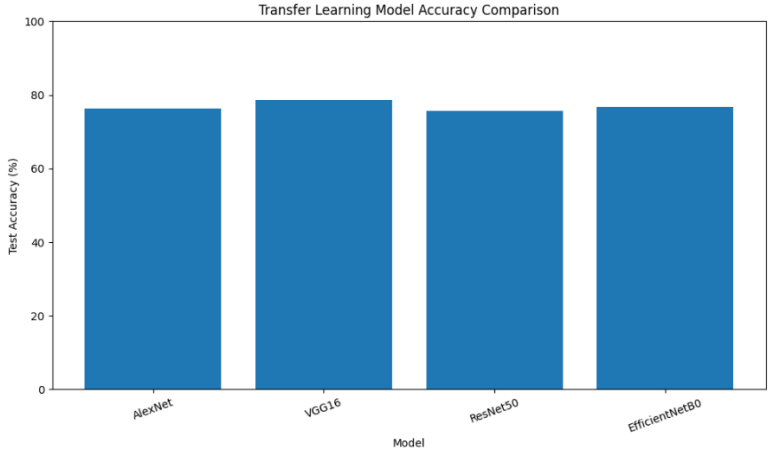
FINAL MODEL COMPARISON

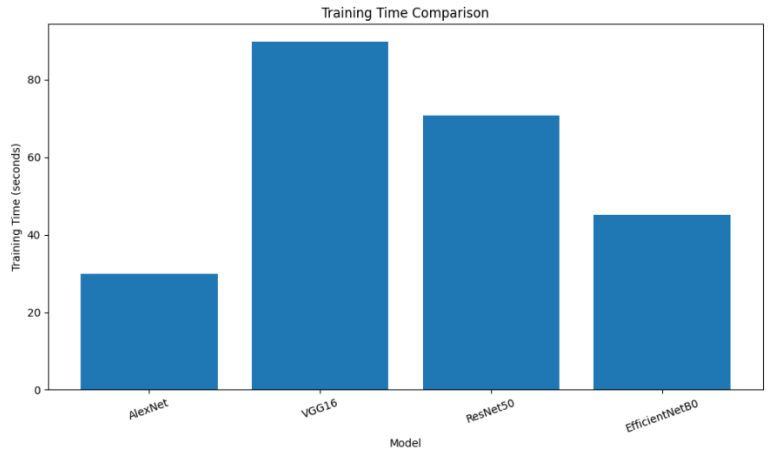
==========

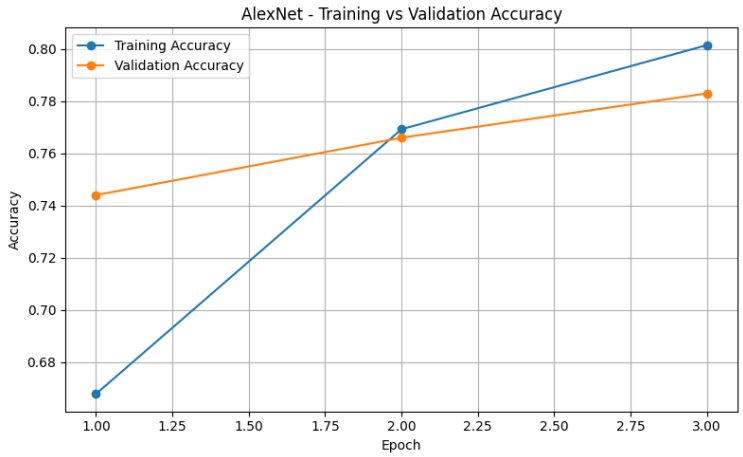
Model	Accuracy (%)	Test Loss	Training Time (sec)
AlexNet	76.2	0.7298	29.8540
VGG16	78.7	0.5900	89.8005
ResNet50	75.6	0.7445	70.7202
EfficientNetB0	76.8	0.7526	45.1065

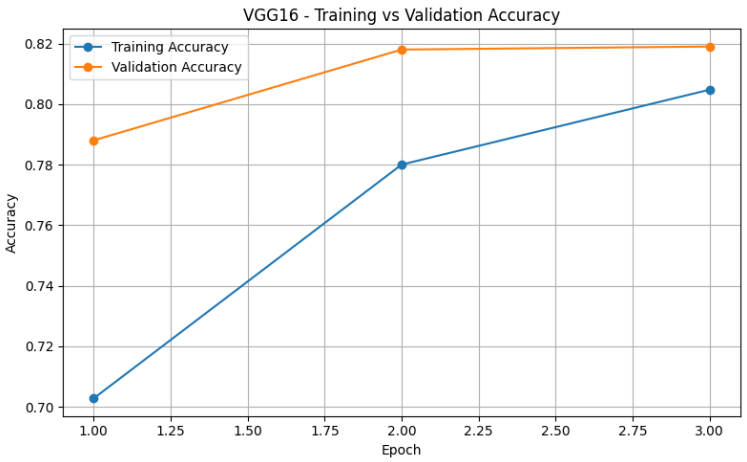
Best Model: VGG16

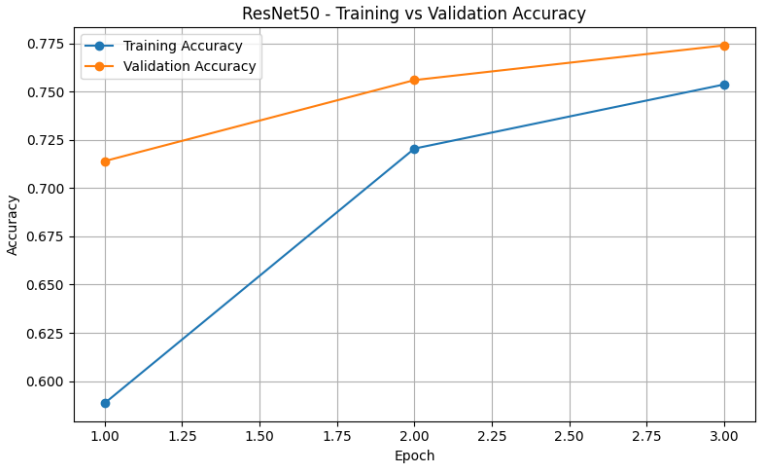
Best Accuracy: 78.70%

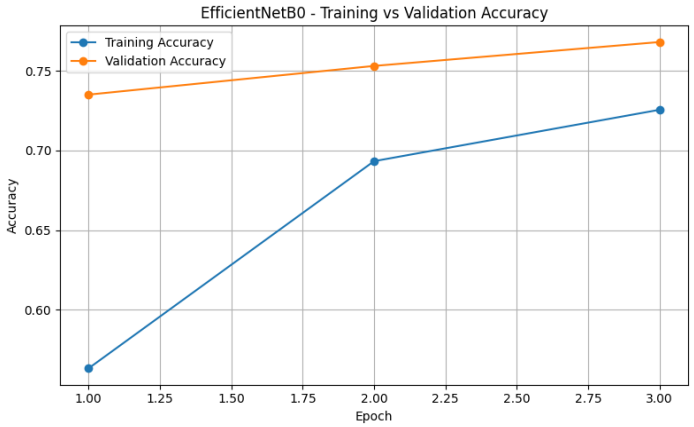


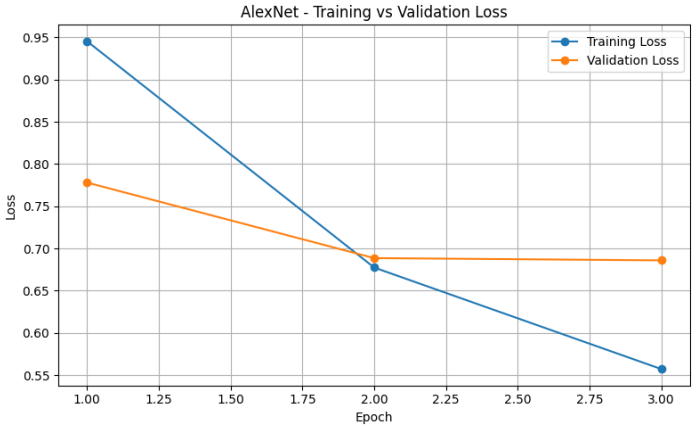


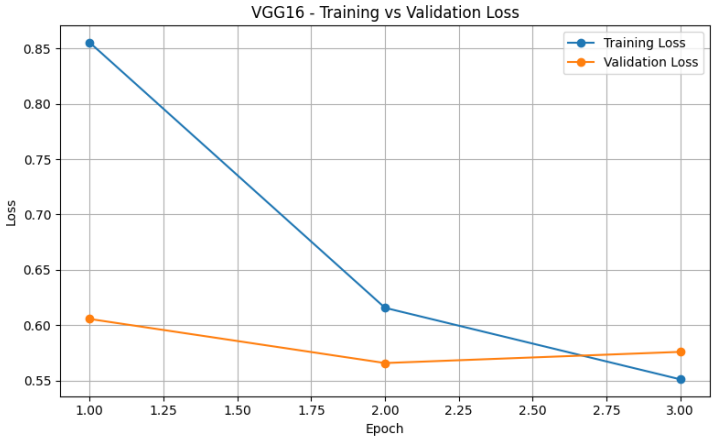

AlexNet - Training vs Validation Accuracy
Training Accuracy
Validation Accuracy
Accuracy
Epoch
0.80
0.78
0.76
0.74
0.72
0.70
0.68
1.00
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00

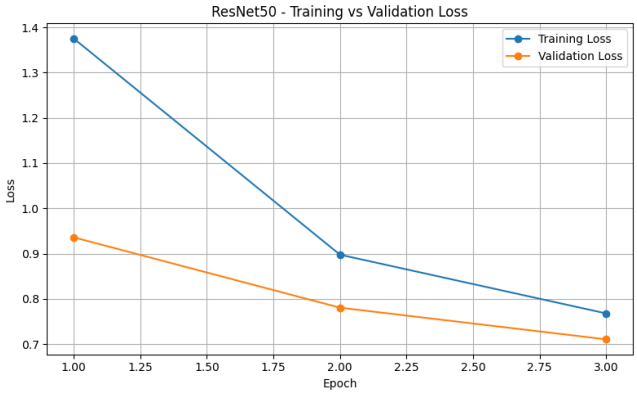

VGG16 - Training vs Validation Accuracy
Training Accuracy
Validation Accuracy
Accuracy
Epoch
0.82
0.80
0.78
0.76
0.74
0.72
0.70
1.00
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00

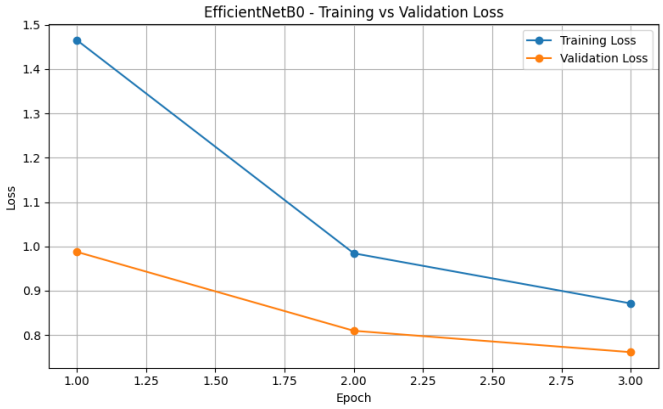

ResNet50 - Training vs Validation Accuracy
Training Accuracy
Validation Accuracy
Accuracy
Epoch
0.775
0.750
0.725
0.700
0.675
0.650
0.625
0.600
1.00
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00

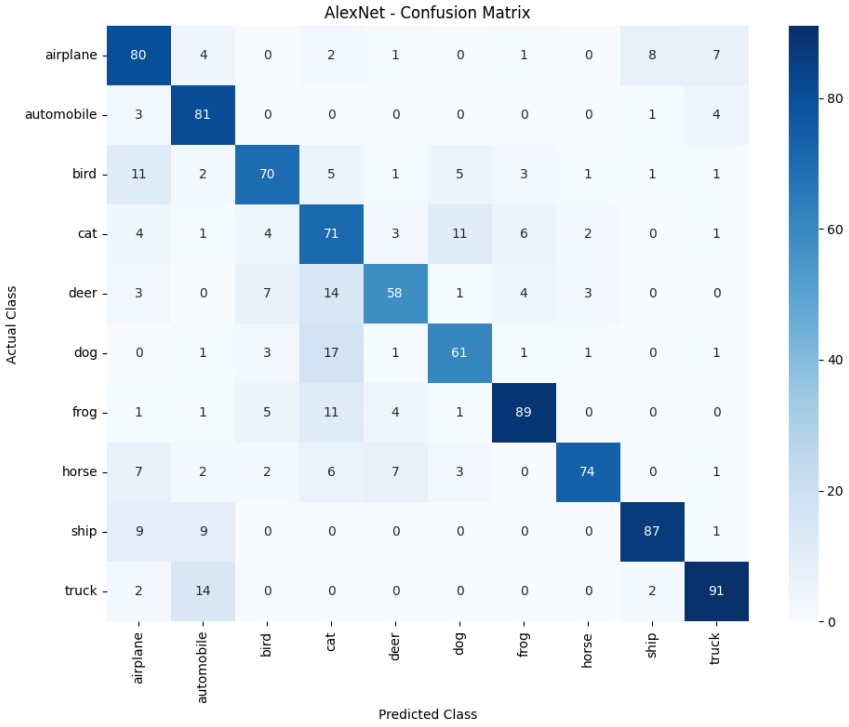

EfficientNetB0 - Training vs Validation Accuracy
Training Accuracy
Validation Accuracy
Accuracy
Epoch
0.75
0.70
0.65
0.60
1.00
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00

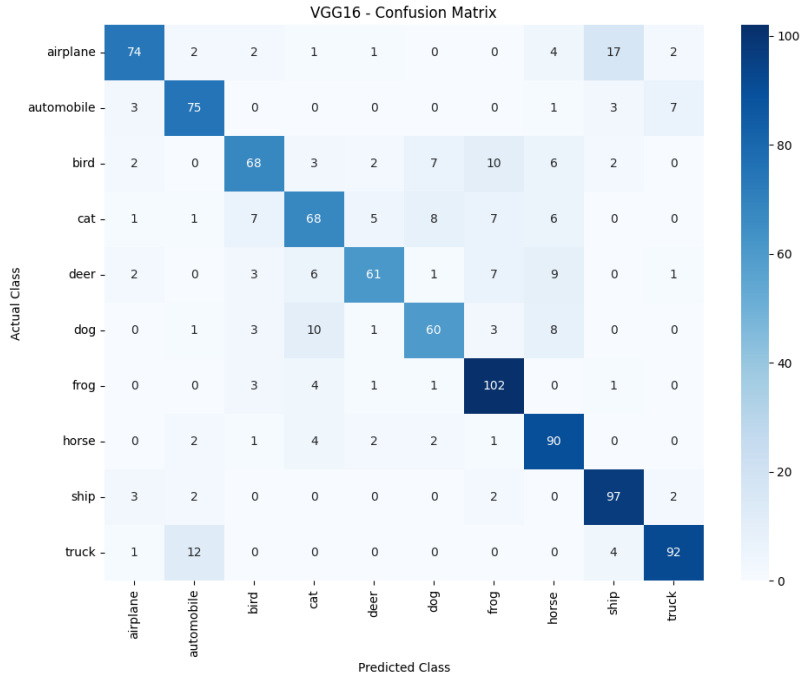

AlexNet - Training vs Validation Loss
Training Loss
Validation Loss
Loss
Epoch
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0.55
1.00
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00

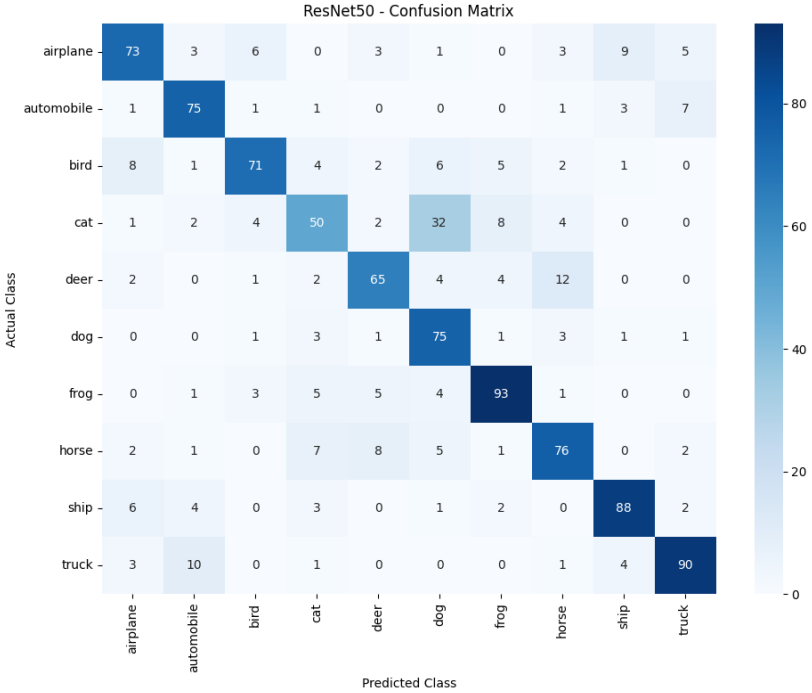

VGG16 - Training vs Validation Loss
Training Loss
Validation Loss
Loss
Epoch
0.85
0.80
0.75
0.70
0.65
0.60
0.55
1.00
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00

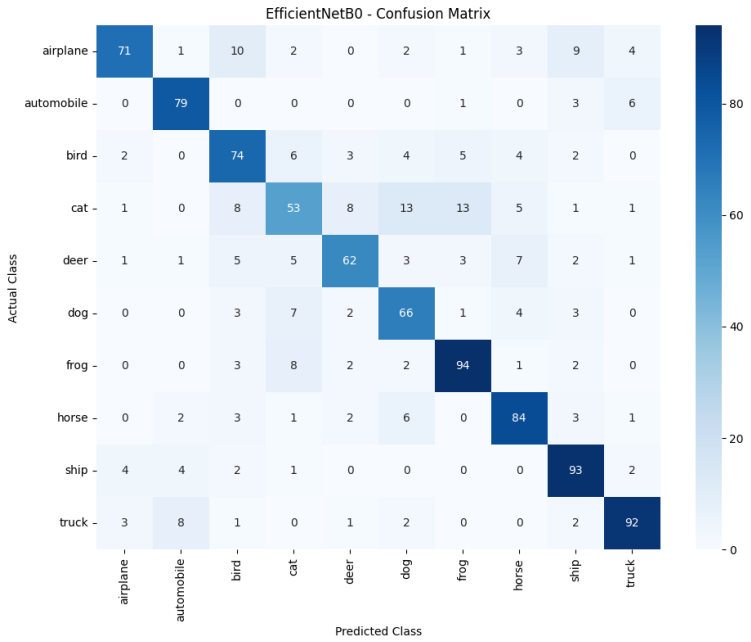

ResNet50 - Training vs Validation Loss
Training Loss
Validation Loss
Loss
Epoch
1.4
1.3
1.2
1.1
1.0
0.9
0.8
0.7
1.00
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00


EfficientNetB0 - Training vs Validation Loss
Training Loss
Validation Loss
Loss
Epoch
1.5
1.4
1.3
1.2
1.1
1.0
0.9
0.8
1.00
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00


AlexNet - Confusion Matrix
Actual Class
Predicted Class
airplane 80 4 0 2 1 0 1 0 8 7
automobile 3 81 0 0 0 0 0 0 1 4
bird 11 2 70 5 1 5 3 1 1 1
cat 4 1 4 71 3 11 6 2 0 1
deer 3 0 7 14 58 1 4 3 0 0
dog 0 1 3 17 1 61 1 1 0 1
frog 1 1 5 11 4 1 89 0 0 0
horse 7 2 2 6 7 3 0 74 0 1
ship 9 9 0 0 0 0 0 0 87 1
truck 2 14 0 0 0 0 0 0 2 91
airplane
automobile
bird
cat
deer
dog
frog
horse
ship
truck
80
60
40
20
0


VGG16 - Confusion Matrix
airplane 74 2 2 1 1 0 0 4 17 2
automobile 3 75 0 0 0 0 0 1 3 7
bird 2 0 68 3 2 7 10 6 2 0
cat 1 1 7 68 5 8 7 6 0 0
deer 2 0 3 6 61 1 7 9 0 1
dog 0 1 3 10 1 60 3 8 0 0
frog 0 0 3 4 1 1 102 0 1 0
horse 0 2 1 4 2 2 1 90 0 0
ship 3 2 0 0 0 0 2 0 97 2
truck 1 12 0 0 0 0 0 0 4 92
Actual Class
airplane automobile bird cat deer dog frog horse ship truck
Predicted Class
100 80 60 40 20 0


ResNet50 - Confusion Matrix
airplane 73 3 6 0 3 1 0 3 9 5
automobile 1 75 1 1 0 0 0 1 3 7
bird 8 1 71 4 2 6 5 2 1 0
cat 1 2 4 50 2 32 8 4 0 0
deer 2 0 1 2 65 4 4 12 0 0
dog 0 0 1 3 1 75 1 3 1 1
frog 0 1 3 5 5 4 93 1 0 0
horse 2 1 0 7 8 5 1 76 0 2
ship 6 4 0 3 0 1 2 0 88 2
truck 3 10 0 1 0 0 0 1 4 90
Actual Class
airplane automobile bird cat deer dog frog horse ship truck
Predicted Class
80 60 40 20 0



CLASSIFICATION REPORT: AlexNet

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	precision	recall	f1-score	support
airplane	0.67	0.78	0.72	103
automobile	0.70	0.91	0.79	89
bird	0.77	0.70	0.73	100
cat	0.56	0.69	0.62	103
deer	0.77	0.64	0.70	90
dog	0.74	0.71	0.73	86
frog	0.86	0.79	0.82	112
horse	0.91	0.73	0.81	102
ship	0.88	0.82	0.85	106
truck	0.85	0.83	0.84	109
accuracy			0.76	1000
macro avg	0.77	0.76	0.76	1000
weighted avg	0.77	0.76	0.76	1000

CLASSIFICATION REPORT: VGG16

==========

	precision	recall	f1-score	support
airplane	0.86	0.72	0.78	103
automobile	0.79	0.84	0.82	89
bird	0.78	0.68	0.73	100
cat	0.71	0.66	0.68	103
deer	0.84	0.68	0.75	90
dog	0.76	0.70	0.73	86
frog	0.77	0.91	0.84	112

horse	0.73	0.88	0.80	102
ship	0.78	0.92	0.84	106
truck	0.88	0.84	0.86	109
accuracy			0.79	1000
macro avg	0.79	0.78	0.78	1000
weighted avg	0.79	0.79	0.78	1000

CLASSIFICATION REPORT: ResNet50

==========

	precision	recall	f1-score	support
airplane	0.76	0.71	0.73	103
automobile	0.77	0.84	0.81	89
bird	0.82	0.71	0.76	100
cat	0.66	0.49	0.56	103
deer	0.76	0.72	0.74	90
dog	0.59	0.87	0.70	86
frog	0.82	0.83	0.82	112
horse	0.74	0.75	0.74	102
ship	0.83	0.83	0.83	106
truck	0.84	0.83	0.83	109
accuracy			0.76	1000
macro avg	0.76	0.76	0.75	1000
weighted avg	0.76	0.76	0.75	1000

CLASSIFICATION REPORT: EfficientNetB0

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	precision	recall	f1-score	support
airplane	0.87	0.69	0.77	103
automobile	0.83	0.89	0.86	89
bird	0.68	0.74	0.71	100
cat	0.64	0.51	0.57	103
deer	0.78	0.69	0.73	90
dog	0.67	0.77	0.72	86
frog	0.80	0.84	0.82	112
horse	0.78	0.82	0.80	102
ship	0.78	0.88	0.82	106
truck	0.86	0.84	0.85	109
accuracy			0.77	1000
macro avg	0.77	0.77	0.76	1000
weighted avg	0.77	0.77	0.77	1000

CONCLUSION

Best performing model: VGG16

Fastest model: AlexNet